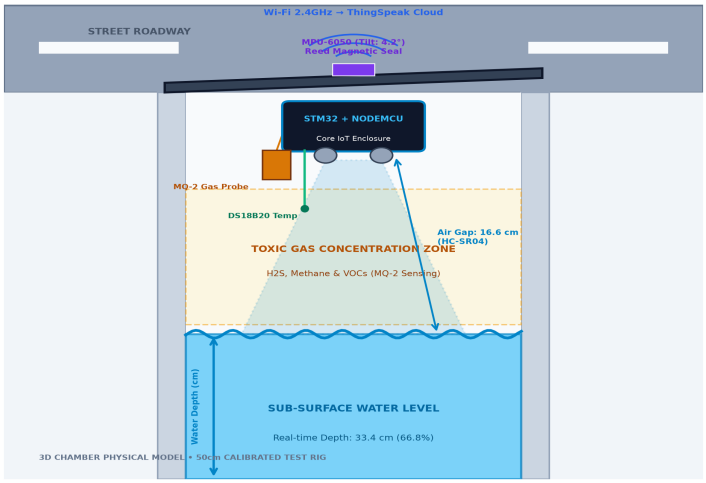
SMART MANHOLE UTILITIES URBAN IOT

Autonomous Sub-Surface Infrastructure Monitoring, Real-Time Cloud Telemetry & 3D Digital Twin SCADA

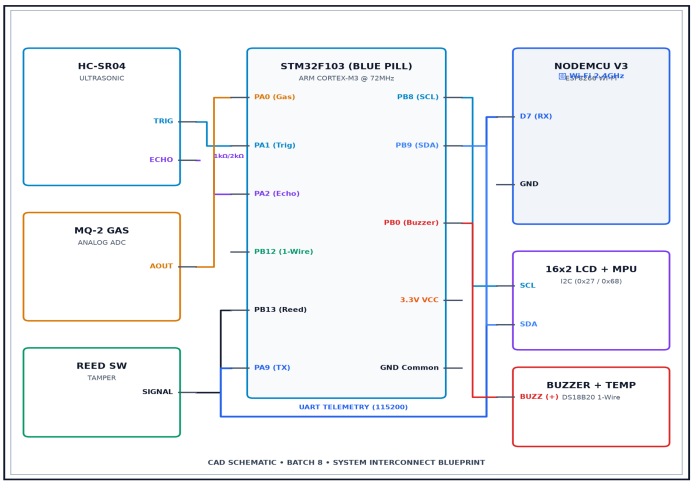
BATCH 8	2400040006 P V Kishor Kumar Reddy	2400040004 T Perumal	2400040303 Chundu Uday Sai	2400040339 Ayan Chowdhury

1. 3D SUB-SURFACE CHAMBER & SENSOR CONE



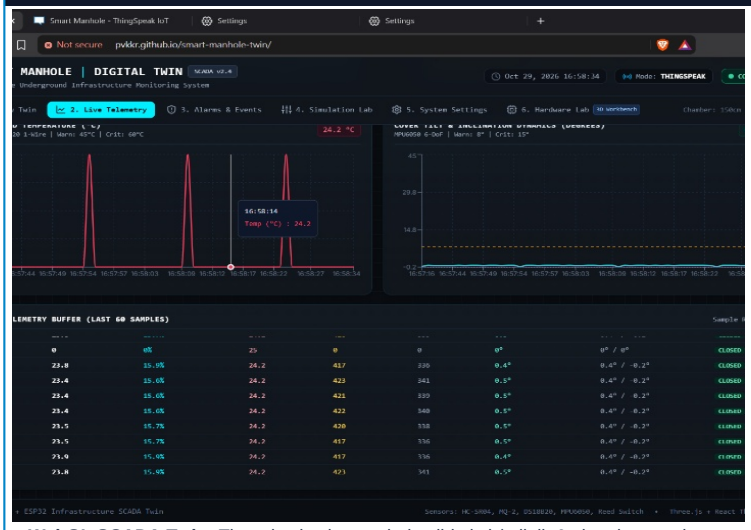
Physical 3D Model: Ultrasonic Beam, Water Level (33.4cm), Gas Zone & Enclosure

2. PROFESSIONAL CAD ELECTRONIC SCHEMATIC



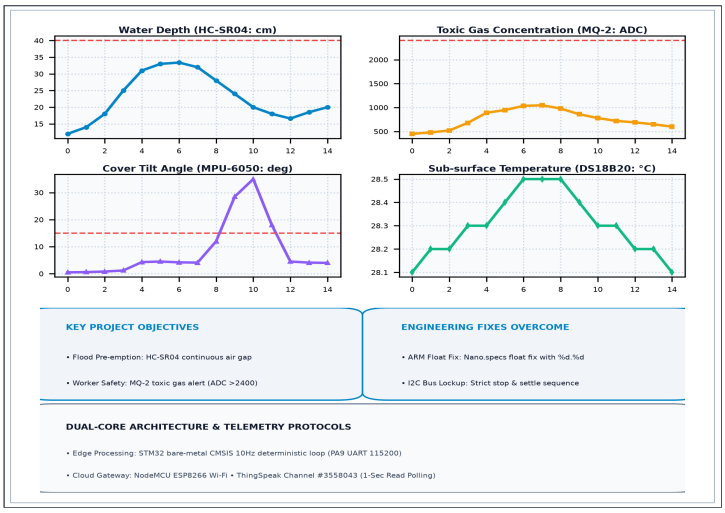
Complete Hardware Bus: Bare-Metal STM32, NodeMCU, 6 Sensors & 1kΩ/2kΩ Divider

3. 3D DIGITAL TWIN SCADA PLATFORM



WebGL SCADA Twin: Three.js physics rendering lid pitch/roll tilt & chamber surcharge

4. LIVE TELEMETRY & FIELD VERIFICATION



Multi-Sensor Analytics: Real-time curves (ThingSpeak #3558043) & telemetry specs

1. PROJECT OBJECTIVES

- **24/7 Multi-Sensor Telemetry:** Autonomous sub-surface monitoring of chamber water depth, sewer gases, tilt & temp.
- **Flash-Flood Pre-emption:** Continuous ultrasonic surcharge tracking alerting city drainage before street-level overflow.
- **Sanitation Worker Safety:** Immediate audio-visual & cloud alarm when toxic gases exceed safe OSHA limits (ADC >2400).
- **Cover Anti-Theft Protection:** Gyroscopic tilt angle detection (>15°) combined with magnetic reed switch seal monitoring.
- **Dual-Mode Zero-Latency Pipeline:** Sub-20ms local REST streaming paired with 1-second ThingSpeak cloud synchronization.

2. REAL-TIME APPS

- **Municipal Flood Management:** Real-time drainage overload telemetry dispatching storm pumps and floodgates automatically.
- **Sanitation Pre-Entry Screening:** Crews verify subterranean air quality before descending, preventing fatal asphyxiation.
- **Urban Road Traffic Safety:** Immediate dispatch alerts when vehicle impacts displace or dislodge heavy cast-iron lids.
- **Critical Infrastructure Defense:** Anti-tamper monitoring guarding underground fiber-optic, power, and gas conduits.
- **Smart City GIS Integration:** Centralized SCADA twin tracking subterranean utility health across municipal grids.

3. BOTTLENECKS SOLVED

- **ARM GCC Nano.specs Float Fix:** Resolved float stripping in embedded snprintf by implementing fixed-point integer math (%d.%d).
- **MPU-6050 I2C Bus Freeze Resolved:** Bit-banged slave ACK lockup eliminated via strict I2C_Stop() + 10µs settle sequence.
- **300s ThingSpeak CDN Latency Solved:** Eliminated browser cache lockup using dynamic timestamp query (?_=Date.now()).
- **STM32 5V Voltage Divider:** 1kΩ/2kΩ passive divider safely stepping 5.0V Echo to 3.33V, safeguarding PA2 pin.
- **Simultaneous 16x2 LCD Character Map:** Formatted all 5 metrics simultaneously (D, G, T, TL, SEAL/OPEN) without screen toggling.

4. FUTURE ROADMAP

- **NB-IoT & LoRaWAN RF Penetration:** Sub-GHz long-range telemetry overcoming RF attenuation through cast-iron utility lids.
- **Piezoelectric Energy Harvesting:** Micro-vibration transducers capturing kinetic mechanical power from traffic vibrations.
- **Edge TinyML Sewage Flow Forecasting:** Onboard neural network detecting flow anomalies and early sewer pipe blockages.
- **Surface-Mounted Solar Studs:** Low-profile roadway solar studs providing perpetual trickle charging for zero-maintenance nodes.
- **Acoustic Wave Depth Profiling:** Multi-frequency ultrasonic depth profiling overcoming sewer steam and foam reflections.